

Appendix 2
Udani DM Checklist
Quality Assessment Checklist for measure the study quality of an ecological systematic review

Category	Item	Yes	No
Reporting	1. Is the objective/aim/hypothesis of the study clearly described?		
	2. Are the main outcomes to be measured clearly described in the Introduction or Methods section? (If the main outcomes are first mentioned in the Results section, the question should be answered “No”)		
	3. Are the characteristics of the bird species included in the study clearly described? In survey study inclusion and exclusion criteria should be given and in case studies, a case-definition and the sources should be given.		
	4. Are the risk factors clearly described?		
	5. Are the conservation methods for specified risk factors clearly described?		
	6. Are the occurrence details (location/time) provided for all bird species considered? (A list of studied bird species is provided)		
	7. Are the main findings of the study clearly described? (Simple outcome data should be reported for all major findings so that the reader can check the major analyses and conclusions. This question does not cover statistical tests which are considered)		
	8. Does the study provide estimates of the random variability in the data for the main outcomes? (In non-normally distributed data the inter-quartile range of results should be reported. In normally distributed data the standard error, standard deviation or confidence intervals should be reported. If the distribution of the data is not described, it must be assumed that the estimates used were appropriate and the question should be answered yes)		
	9. Have actual probability values been reported (e.g. 0.045 rather than <0.05) for the main outcomes except where the probability value is less than 0.001?		
External validity (All the following criteria attempt to address the findings of the study and whether they may be generalised to the bird population from which the study data were derived.)	10. Does the study identify the sources of the data, describing how the data were collected clearly? Bird species would be representative if they comprised the entire source bird population, an unselected a random sample. Random sampling is only feasible where a list of all members of the relevant population exists. Where a study does not report the proportion of the source population from which the bird species are derived, the question should be answered as unable to determine)		
	11. Does the duration of data which had taken from the data source clearly stated?		
	12. Was the data collected sites clearly stated?		
	13. Were the data obtained from a reputable source which validates data appropriately?		
Internal validity - bias	14. Is autocorrelation associated with time or spatial variation considered in the analysis?		
	15. Does the study report any retrospective unplanned subgroup analyses?		
	16. Were the statistical tests used to assess the main outcomes appropriate? The statistical techniques used must be appropriate to the data. For example, nonparametric methods should be used for small sample sizes. Where little statistical analysis has been undertaken but where there is no evidence of bias, the question should be answered yes. If the distribution of the data (normal or not) is not described it must be assumed that the estimates used were appropriate and the question should be answered yes.		
Internal validity - confounding (selection bias)	17. Were the main outcome measures used accurate (valid and reliable)? For studies where the outcome measures are clearly described, the question should be answered yes. For studies which refer to other work or that demonstrates the outcome measures are accurate, the question should be answered as yes.		
	18. Was this a comparative study?		
	19. For comparative studies was the data collection effort, duration and area are the same/similar?		
	20. For comparative studies was the same data collection method used?		
Power	21. In comparative studies were an attempt made to adjust for any differences in data collection effort or data collection method?		
	22. For comparative studies was sufficient data collected (over space and time) in order to allow testing with appropriate levels of significance (5%) and power (80%)? (Answer No if this issue was not addressed)		

Udani DM scores have been categorized using quartile mean scores (Poor, Fair, Average, Good and Excellent)